

PRATYUSH AVI

(480)-401-7629 | pratyushavi@nyu.edu | [Website](#) | [GitHub](#)

EDUCATION

New York University

September 2023 – May 2028 (expected)

PhD, Computer Science

New York, NY

Prospective Thesis Focus: Applying techniques from theoretical computer science and random numerical linear algebra for practically faster and provably efficient algorithms for problems in scientific machine learning and approximate nearest neighbor search.

I am currently supported by the Department of Education GAANN Fellowship.

Arizona State University

August 2019 – December 2022

BSE, Computer Systems Engineering & Barrett, The Honors College

Tempe, AZ

GPA: 4.00

PUBLICATIONS

Query Efficient Structured Matrix Learning

Noah Amsel, **P. Avi**, Tyler Chen, Feyza Duman Keles, Chinmay Hegde, Christopher Musco, Cameron Musco, David Persson
Conference on Learning Theory 2026, arXiv:2507.19290

Preliminary version at *Theory of Scientific Computing Workshop* at COLT 2025

Poster presentations at COLT 2025 and Simons Institute Workshop on Linear Systems and Eigenvalue Problems 2025

EXPERIENCE

PhD Research Intern

May 2023 – August 2023

MIT Lincoln Laboratory, Group 33 - Advanced Sensor Systems & Test Beds

Lexington, MA

- Built a real-time planning web interface using Python and React.js, improving user interface usability and deployment for mission operators.
- Engineered a domain-specific planning language to automate real-time mission execution and reduce operator workload.
- Mentored junior interns, providing technical guidance to support project completion.

Undergraduate Research Intern

May 2022 – August 2022

MIT Lincoln Laboratory, Group 33 - Advanced Sensor Systems & Test Beds

Lexington, MA

- Built scalable machine learning and signal processing models (CUDA, PyTorch) to detect key events in large-scale radar datasets.
- Implemented YOLOv3 for spectrogram-based signal classification and prototyped a graph neural network model that used raw-spectrogram data for signal recognition.
- Developed a user interface in React.js for a secure meeting recording and transcription utility.
- Gained hands-on radar experience through field-deployment project.

Machine Learning Undergraduate Researcher

May 2021 – April 2022

Purdue University & U.S. Army

West Lafayette, IN

- Built a graph neural network in PyTorch to predict properties of organic molecules relevant to energetics and material science research.
- Wrote a Jupyter Notebook based teaching utility to teach machine learning approaches to undergraduate students.
- Paper Link: [Machine Learning for Data Driven Energetics Materials Screening and Performance Design](#)
- Presented findings to leading energetics and Army Research Lab researchers.

PROJECTS

Multi-Scale Visualization of Cellular Features in Cancerous Image Data | *Deep Learning, Data Visualization*

- Used Vision Transformer based encoders to generate high-dimensional embeddings for individual cells in a collection of multiplexed cancer image data. This is combined with a high-fidelity visualization tool to discover and explore novel features in the medical samples.

Dimensionality Reduction for Spatial Audio | *Artificial Intelligence, Data Science, Extended Reality*

- Used random binary matrices for compressing Head-Related-Transfer-Functions for spatial audio
- Project Report Link: [Random Projections for Dimensionality Reduction of HRTFs](#)

Evaluating Hierarchical Learning using MAXQ and Q-Learning | *Artificial Intelligence, Reinforcement Learning*

- Re-implemented and evaluated hierarchical learning by comparing MAXQ and Q-Learning in the OpenAI Gym.
- Project Report Link: [Re-implementation of MAXQ with the OpenAI Gym in the Taxi Domain](#)

TEACHING

Teaching Assistant

New York University

January 2026 – May 2026

New York, NY

- **Course:** Algorithmic Machine Learning and Data Science with Prof. Ainesh Bakshi
- [Course Website](#)

Teaching Assistant

New York University

January 2025 – May 2025

New York, NY

- **Course:** Algorithmic Machine Learning and Data Science with Prof. Chris Musco
- [Course Website](#)

Teaching Assistant

Arizona State University

January 2021 – December 2022

Tempe, AZ

- **Course:** Introduction to Programming Languages with Prof. Justin Selgrad

AWARDS & HONORS

Best Poster

MIT Lincoln Laboratory

July 2022

Lexington, MA

- Led finalist team at MIT LL's Intern Innovation Idea Challenge, pitching a novel deep-sea exploration vehicle concept.
- Conceptualized vehicle design using MIT LL technologies, along with a supporting research and development pipeline.

Dean's List

Arizona State University

August 2019 - December 2022

Tempe, AZ

- All semesters through Fall 2019 - Fall 2022

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Dart, MySQL, JavaScript, CSS/SASS, HTML, Julia, Verilog, RISC-V

Frameworks: AWS, PyTorch, Tensorflow, Jupyter, CUDA, Vue.js, d3.js, Flutter, React.js, Electron.js

Developer Tools: Linux/UNIX, Intel Pin, Arduino, Raspberry Pi, Git, Firebase, MATLAB